

TP5

Image Segmentation

The objective of this practical work is to study image segmentation through the K-means method.

1 The method

K-means consists in finding regions in the image that minimize the following energy:

$$F(\text{regions}, \text{pixels}) = \sum_{i \in \text{regions}} \sum_{j \in \text{region } i} (x_j - c_i)^t (x_j - c_i),$$

where x_j is the value taken into account at pixel j and c_i is the value of the center of region i . Starting from an initial solution, the principle is to iterate 2 steps:

1. Assuming known region centers, associate each pixel to its closest region (i.e. with the minimum distance to the region center).
2. Assuming known associations between pixel and regions, determine new region centers as the mean of pixel values in each region.

2 Implementation

Check `kmeans.c` which implements the main program. Then complete the function `kmeans()` in `TP5/TP5_func.c`:

1. Implement K-means with x values being the RGB values in the image. Test it with

```
./kmeans images/frog.ppm -1 5 0 0 0 0
```

Now explore using different input values:

2. What is the influence of the number of regions K ?
3. What is the influence of the initial values for region centers ?

4. Which stop condition did you use? Propose a second one.

Consider now for x values both RGB and location (i,j) in the image. Use the `location` argument to activate this option.

5. How does it change the results ?
6. How can we balance the influence of colors and locations in the image? Use the `location` argument to have more control over the balance.

Now that you have the code and understood the role of the parameters

7. Use a different image from frog and explain how you chose K, the initial values and the location-color balance to get what you consider is a good segmentation result.

3 Bonus

1. Save the cluster structure into a CSV file as well as an image with the value of each cluster index in each pixel (mask.pgm).
2. Modify the function `ppmtopam` from TP1 by adding an integer value as input: it defines which value of the mask should be used for 255 opacity and the rest as 0 opacity.
3. Use any online converter to generate a PNG file from the pam file. You can segment any image and use it as a sticker in your favorite application!